

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and **COURSE CODE:** CSA15
Big Data Analytics

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct

I Yarra Pawan (192424294) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Y.Pawan

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

During major seasonal sales such as Amazon Great Indian Festival, Flipkart Big Billion Days, or Myntra End of Reason Sale, website traffic can increase from thousands to millions of users within minutes. If the company uses a fixed number of servers, they may become overloaded, causing slow performance, payment failures, or even website crashes. Cloud computing addresses this problem through auto-scaling, load balancing, and monitoring.

Auto-Scaling

Cloud auto-scaling automatically increases or decreases the number of virtual machines based on demand. When CPU usage, memory usage, or incoming requests exceed a predefined limit, additional servers are launched automatically. Once traffic decreases, the extra servers are removed to reduce costs.

For example, an e-commerce website normally running on 10 virtual machines can automatically scale up to 100 virtual machines during a flash sale and scale back after the event.

Load Balancing

Load balancing distributes customer requests evenly across all available servers instead of sending them to a single server. It also checks server health and redirects traffic to healthy servers if one fails.

Benefits:

- Prevents server overload
- Improves response time
- Increases availability
- Supports fault tolerance
- Ensures uninterrupted service

Reducing Downtime and Improving User Experience

Together, auto-scaling and load balancing ensure enough resources are available during peak demand. Customers benefit from:

- Faster website loading
- Smooth product browsing
- Successful checkout and payments
- Continuous service availability

This improves customer satisfaction and increases business revenue.

Role of Monitoring Tools

Monitoring tools track system performance by collecting metrics such as:

- CPU and memory usage
- Network traffic
- Response time
- User requests
- Application errors

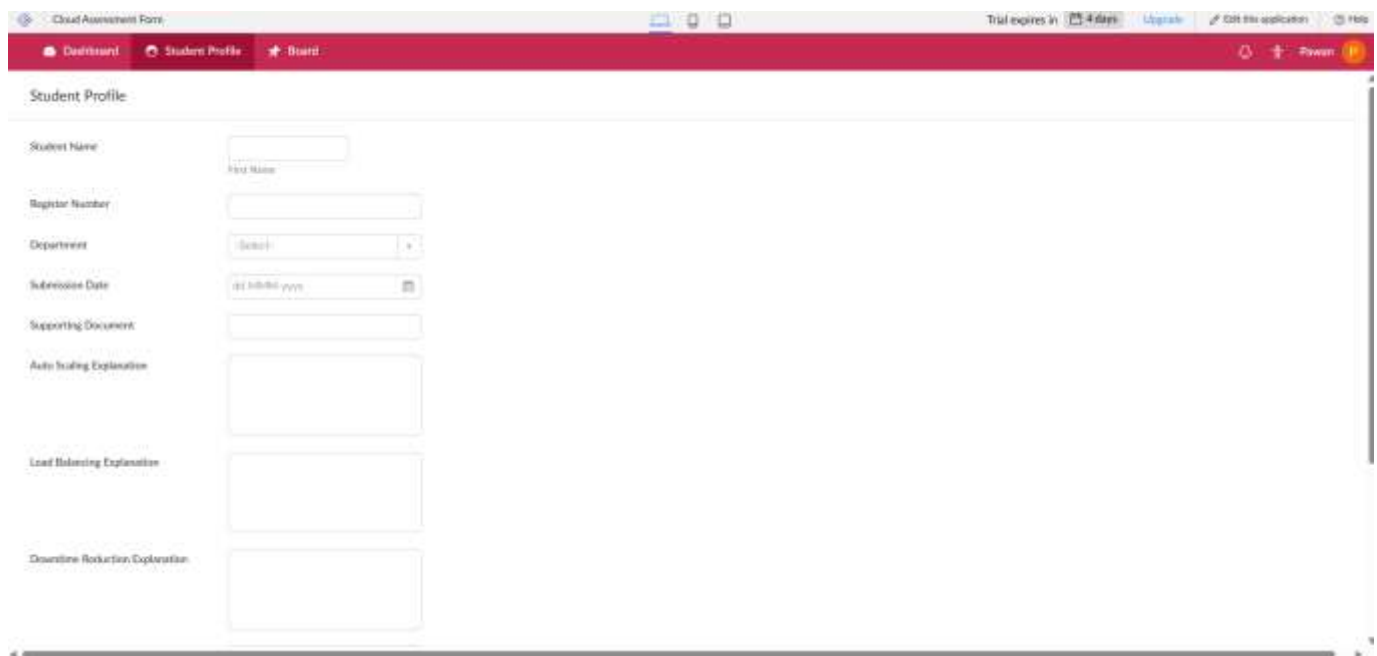
Services like Amazon CloudWatch, Azure Monitor, and Google Cloud Monitoring generate alerts, trigger auto-scaling, and help administrators detect bottlenecks and optimize performance.

Real-World Example

During Amazon Prime Day, AWS Auto Scaling launches additional EC2 instances as traffic increases. Elastic Load Balancer distributes requests across servers, while Amazon CloudWatch monitors system health and triggers scaling automatically. This ensures fast page loading and successful transactions even during peak traffic.

Conclusion

Auto-scaling, load balancing, and monitoring together provide a scalable, reliable, and highly available cloud infrastructure. They minimize downtime, improve user experience, optimize costs, and help businesses efficiently manage sudden traffic spikes.



The screenshot displays a web application interface for a 'Cloud Assessment Form'. The top navigation bar is red and contains links for 'Dashboard', 'Student Profile', and 'Board'. A trial expiration notice 'Trial expires in 4 days' is visible in the top right corner. The main content area is titled 'Student Profile' and contains several form fields: 'Student Name' (with a sub-field for 'First Name'), 'Register Number', 'Department' (a dropdown menu), 'Submission Date' (a date picker), 'Supporting Document', 'Auto Scaling Explanation', 'Load Balancing Explanation', and 'Downtime Reduction Explanation'. Each of the last three explanation fields is accompanied by a large text area for input.

<

>

Edit

Duplicate

More ▾

×

Student Name	Eva Garcia
Register Number	STU2026005
Department	Mechanical
Submission Date	05-Aug-2026
Supporting Document	Terraform Guide.pdf
Auto Scaling Explanation	Infrastructure as code enables rapid scaling
Load Balancing Explanation	DNS load balancing routes users globally
Downtime Reduction Explanation	Immutable infrastructure ensures zero downtime
Cloud Monitoring Explanation	Datadog provides comprehensive observability
E-commerce Example	Prime Day inventory scaling

 Add a comment

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Organizations generate large amounts of business data, including customer records, financial transactions, employee information, and application databases. Losing this data because of hardware failures, cyberattacks, or natural disasters can seriously affect business operations. Cloud storage provides a secure, durable, and cost-effective backup solution.

Durability

Cloud providers such as Amazon S3, Microsoft Azure Blob Storage, and Google Cloud Storage offer extremely high durability (up to 99.999999999%, or 11 nines) by storing multiple copies of data across different storage devices and data centers. If one storage device fails, other copies remain available.

Data Redundancy

Cloud providers automatically create multiple replicas of backup data using:

- Local redundancy
- Zone redundancy
- Regional redundancy
- Cross-region replication

This ensures data remains available even if an entire data center becomes unavailable.

Encryption

Encryption converts readable data into unreadable ciphertext.

- Data at Rest: Protected using AES-256 encryption.
- Data in Transit: Secured using TLS/SSL encryption.

This prevents unauthorized users from accessing the data even if it is intercepted.

Access Control

Access control ensures only authorized users can access backup data through:

- Identity and Access Management (IAM)
- Role-Based Access Control (RBAC)

- Multi-Factor Authentication (MFA)
- Password policies
- Audit logs

For example, only backup administrators can delete backup files, while employees have read-only access.

Cost Benefits

Cloud storage offers:

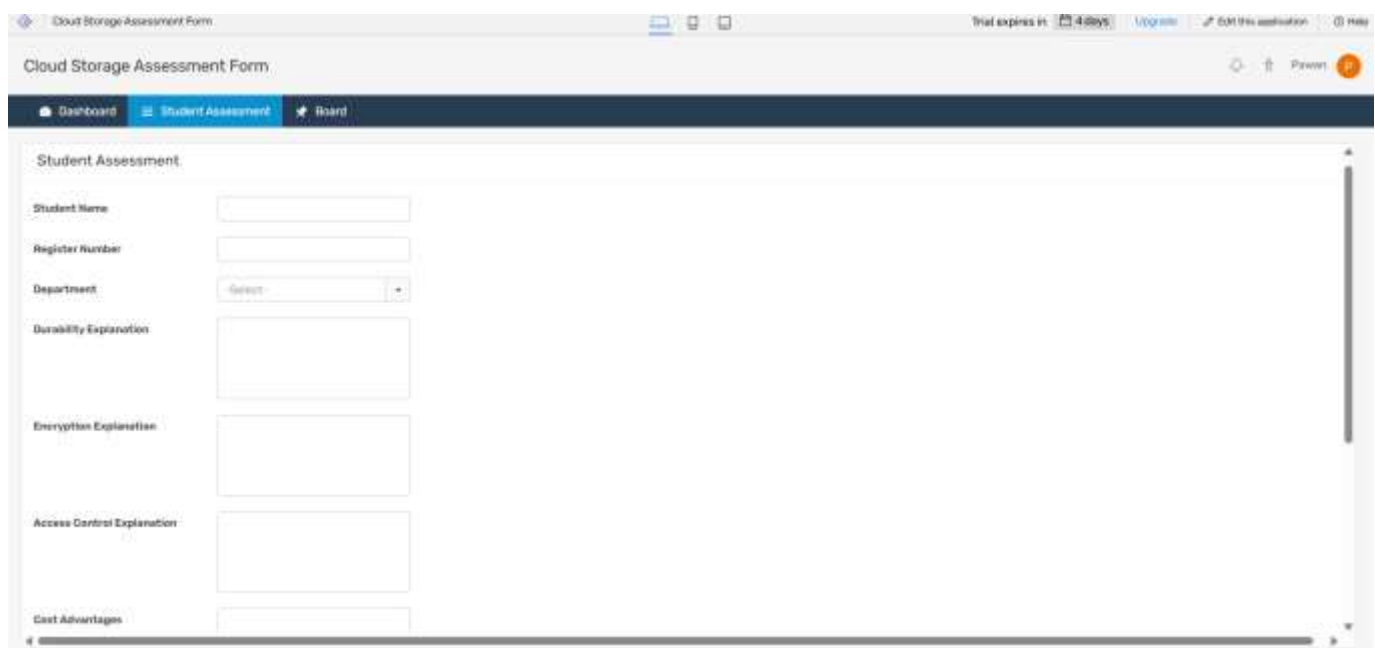
- Pay only for storage used
- No hardware purchase
- Automatic maintenance
- Easy scalability
- Lower operational costs

Disaster Recovery Example

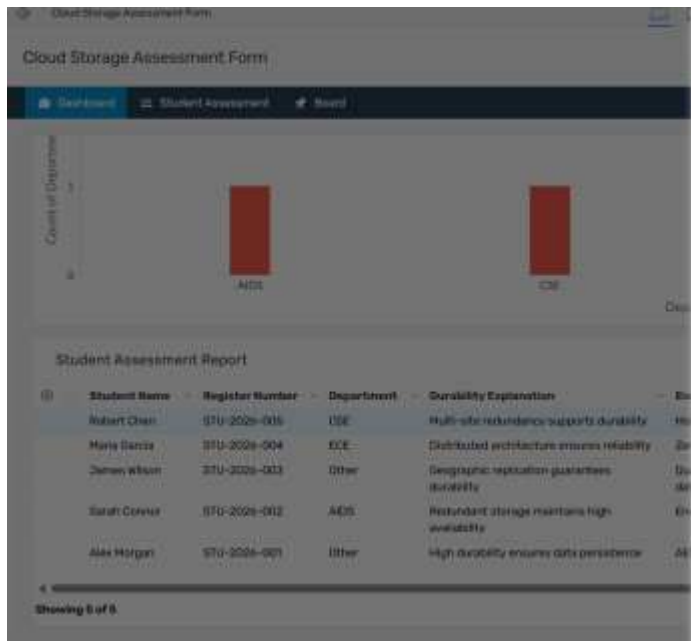
A hospital stores patient records in Amazon S3 with cross-region replication. When a flood damages its primary data center, administrators restore the encrypted backups from another AWS region, allowing hospital services to resume quickly without losing critical patient data.

Conclusion

Cloud storage provides secure, durable, scalable, and highly available backup solutions. Features such as redundancy, encryption, IAM, and disaster recovery make it more reliable and cost-effective than traditional backup systems.



The screenshot shows a web application titled "Cloud Storage Assessment Form". The interface includes a top navigation bar with a "Dashboard" link, a "Student Assessment" link (which is active), and a "Board" link. Below the navigation bar, the "Student Assessment" form is displayed. It contains several input fields: "Student Name", "Register Number", "Department" (a dropdown menu), "Durability Explanation", "Encryption Explanation", "Access Control Explanation", and "Cost Advantages". The form is set against a light gray background with a dark blue header. In the top right corner of the browser window, there is a trial expiration notice: "Trial expires in 4 days" and a "Log out" button.



Overview

Student Name	Robert Chan
Register Number	STU-2026-005
Department	CSE
Durability Explanation	Multi-site redundancy supports durability
Encryption Explanation	Homomorphic encryption allows computation
Access Control Explanation	Dynamic access policies adapt to needs
Cost Advantages	Subscription tiers offer flexibility
Disaster Recovery Case	Instant recovery reduces recovery time
Supporting Document	
Submission Date	06-Aug-2026

[Add a comment](#)

Cloud Storage Assessment Form

Student Assessment

Basic Fields

Name Email

Address Phone

Single Line Multi Line

URL

Number Date

Time Drop Down

Radio Multi Select

Checkbox

Field Properties

Field Name

Field Link Name

Field Link Reference

Validation

Mandatory Non-duplicate values

Initial Value

Initial Value

Character Maximum

255

Data Privacy

Contains personal data (PII)

Data Security

Encrypt data

Protects health information (PHI)

User Input (for mobile)

QR Code Barcode

Tridacteam is a leading provider of cloud storage solutions. | [Helpdesk](#) | [Contact Us](#)

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

A mobile application serving users worldwide must deliver content quickly regardless of location. Without optimization, users far from the main server experience high latency, slow loading, and poor performance. Content Delivery Networks (CDNs) solve this problem by caching content closer to users

Content Delivery Network (CDN)

A CDN is a globally distributed network of servers that stores cached copies of frequently accessed content such as images, videos, JavaScript files, and application updates. Instead of downloading content from the origin server every time, users receive it from the nearest CDN server.

Edge Servers

Edge servers are CDN servers located in different geographic regions. Their benefits include:

- Lower latency
- Faster loading
- Reduced bandwidth usage
- Improved availability
- Better scalability

For example, a user in Chennai receives content from a nearby Indian edge server instead of a server in the United States.

Global Availability

Cloud providers maintain global networks of regions, availability zones, and CDN edge locations.

Examples include:

- AWS CloudFront
- Azure CDN
- Google Cloud CDN

These services automatically route users to the nearest healthy server, ensuring reliable performance even during regional outages.

Real-Time Streaming Example

Netflix stores movies in cloud storage while AWS CloudFront delivers video content through edge servers worldwide. A customer in India receives video from the nearest edge server, resulting in:

- Faster playback
- Reduced buffering
- High-definition streaming
- Better user experience

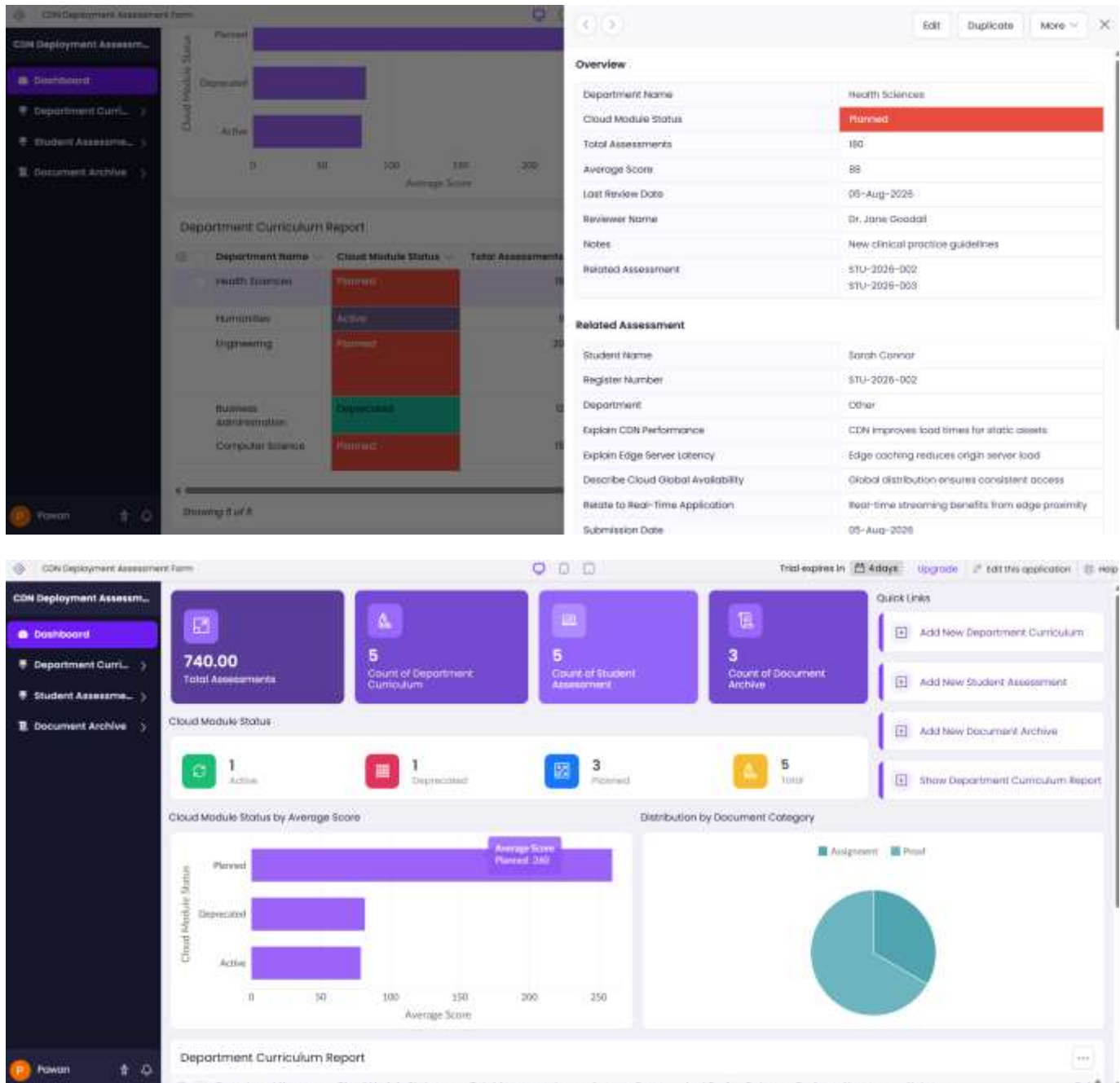
Gaming Example

Online games such as PUBG, Valorant, and Fortnite use edge servers to reduce latency. Players connect to nearby servers, improving response time and providing smoother gameplay.

Conclusion

CDNs improve application performance by reducing latency, increasing availability, and delivering content faster through geographically distributed edge servers, ensuring a better experience for users worldwide.

The screenshot shows a form builder interface for a 'CDN Placement Assessment' form. The form is titled 'CDN Placement Assessment' and 'Department Contribution'. It features a 'Basic Fields' sidebar on the left with various field types like Name, Email, Address, Phone, Single Line, Multi Line, Y2I, Number, Date, Time, Drop Down, Radio, Multi Select, and Check Box. The main form area contains several fields: 'Department Name' (text), 'Cloud Mobile Email' (text), 'Total Assessments' (text), 'Package Score' (text), 'Last Review Date' (text), 'Reviewer Name' (text), 'Notes' (text), and 'Related Assessment' (text). A 'Field Properties' sidebar on the right allows for configuring each field, including setting the field name, label, validation rules (mandatory, initial value, character maximum), data privacy (contains personal data), data security (encrypt data, contains health info), and user input (for mobile, create, QR code, barcode). The form is currently in 'Draft' status.



4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Financial institutions handle highly sensitive data such as customer account details, credit card information, loan records, and transaction histories. They must comply with regulations like PCI DSS, ISO 27001, SOC 2, and RBI guidelines. Cloud providers offer security services and compliance tools to help organizations meet these requirements.

Regulatory Compliance and Auditing

Cloud providers undergo regular independent audits and provide compliance certifications. They also offer tools to maintain audit trails and enforce security policies.

Examples include:

- AWS Artifact
- Azure Compliance Manager
- Google Cloud Compliance Center

These services provide reports and documentation required during regulatory audits.

Logging

Logging records important activities such as:

- User logins
- File access
- Configuration changes
- API calls
- Financial transactions

Services like AWS CloudTrail and Azure Activity Log create audit trails that help investigate incidents and demonstrate compliance.

Monitoring

Monitoring continuously tracks system health and detects suspicious activities such as:

- Unusual login attempts
- High network traffic
- Failed authentication
- Unauthorized access

Tools like Amazon CloudWatch, Azure Monitor, and Google Cloud Monitoring generate alerts for quick response.

Encryption

Encryption protects sensitive financial data:

- Data at Rest: AES-256 encryption secures stored databases and backups.

- Data in Transit: TLS/SSL encrypts communication between users and cloud servers.

This ensures intercepted data cannot be read without the correct encryption keys.

Data Sovereignty

Cloud providers allow organizations to store data in specific geographic regions to meet local regulations. For example, an Indian bank can store customer data in the AWS Mumbai Region or Azure Central India to comply with RBI data localization requirements.

Real-World Example

A bank providing online fund transfers uses TLS for secure data transmission and AES-256 for data storage. AWS CloudTrail records all API activities, CloudWatch monitors suspicious logins, and IAM restricts access to authorized employees. Customer data is stored in the AWS Mumbai Region to meet compliance requirements.

Conclusion

Cloud providers help financial institutions achieve regulatory compliance through auditing, logging, monitoring, encryption, and regional data storage. These features improve security, support compliance, and protect sensitive financial information.

The screenshot displays a 'Cloud Security Assessment Form' interface. On the left, a 'Student Assessment Report' table lists five students with their details and compliance explanations. On the right, an 'Overview' table provides a summary of key security measures. The interface includes navigation buttons at the top and a 'Showing 5 of 5' indicator at the bottom left.

Student Name	Register Number	Department	Compliance Explanation
Robert Chen	STU-2028-008	AML	Automated vulnerability scanning active daily
Maria Garcia	STU-2028-004	FSE	Zero-trust architecture fully implemented
James Wilson	STU-2028-013	ESE	Regular penetration testing conducted monthly
Sarah Connor	STU-2028-010	Other	Multi-factor authentication enforced for all users
Alex Morgan	STU-2028-009	AES	Data encrypted at rest and in transit

Overview	
Student Name	Robert Chen
Register Number	STU-2028-008
Department	AML
Compliance Explanation	Automated vulnerability scanning active daily
Monitoring Role	DevSecOps Lead
Encryption Protection	Quantum-resistant crypto
Sovereignty Assurance	Hybrid Cloud Setup
Financial Transaction Example	Tokenized payment processing
Submission Date	06-Aug-2028
Supporting Document	

Cloud Security Assessment Form

Trial expires in 4 days | [Log out](#) | [Edit this application](#) | [Help](#)

Dashboard | Student Assessment | Board

Student Assessment

Student Name

Register Number

Department

-Select-

Compliance Explanation

Monitoring Role

Encryption Protection

Severance Insurance

Cloud Security Assessment Form

Trial expires in 4 days | [Log out](#) | [Edit this application](#) | [Help](#)

Dashboard | Student Assessment | Board

Board

Search

+

...

Civil	CSE	EEE	ECE	Mechanical
No Records in Civil	No Records in CSE	James Wilson 0010-0000-0000	Maria Garcia 0010-0000-0000	No Records in Mechanical

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Modern applications require fast deployment, efficient resource usage, and easy scalability. Containerization packages an application with its dependencies so it runs consistently across different environments. Docker is used to create containers, while Kubernetes manages them.

Containers vs. Virtual Machines

Virtual Machines (VMs):

- Include a full guest OS
- Use more CPU, memory, and storage
- Slower to start

Containers:

- Share the host OS kernel
- Use fewer resources
- Start quickly
- Support higher application density

Kubernetes

Kubernetes automates container management by providing:

- Automatic deployment
- Auto-scaling
- Load balancing
- Self-healing
- Rolling updates

These features improve availability and reduce manual effort.

Portability

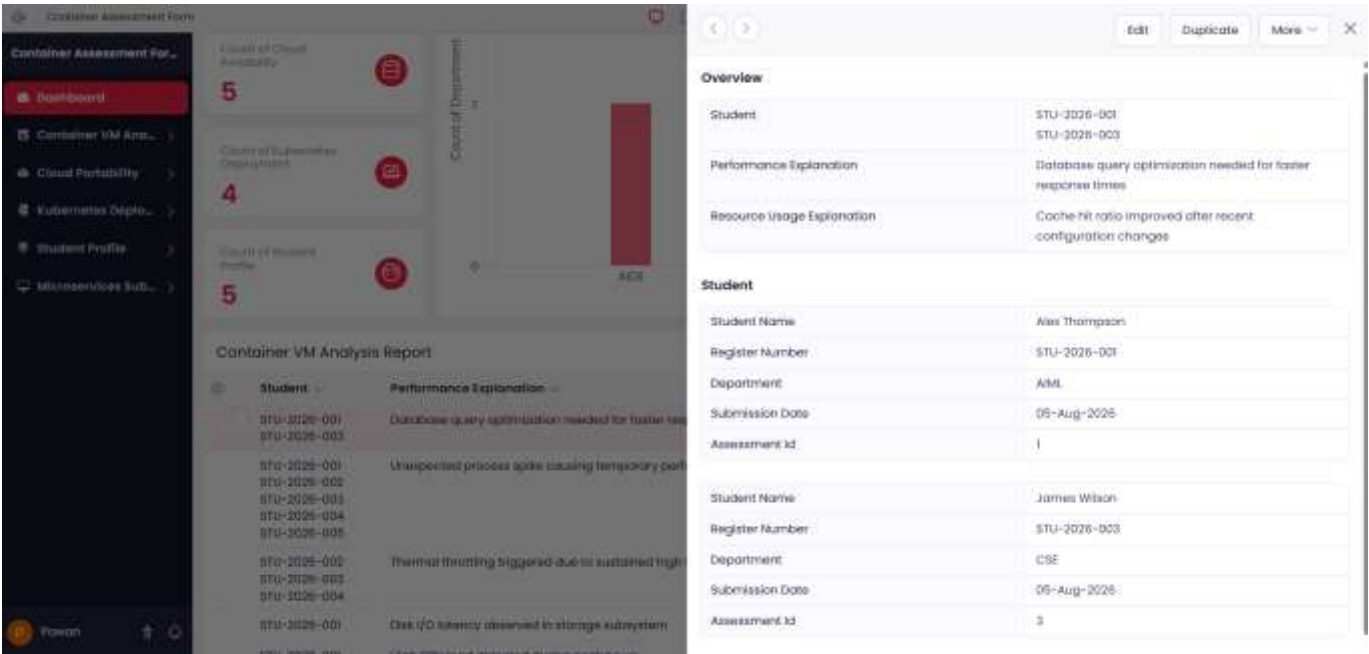
Containers can run on AWS (EKS), Azure (AKS), Google Cloud (GKE), or on-premises Kubernetes clusters, making applications portable and reducing vendor lock-in.

Example

A food delivery app like Swiggy uses separate containers for services such as Authentication, Orders, Payments, and Delivery. During peak hours, Kubernetes automatically adds more Order Service containers and replaces failed containers without affecting other services.

Conclusion

Containerization provides fast deployment, efficient resource usage, and portability, while Kubernetes ensures automation, scalability, and high availability, making them ideal for modern cloud applications.



Container Assessment Form

Trid expires in 4 days Upgrade Edit this application Help

Container Assessment For...

Dashboard

Container VM Ana...

Cloud Portability

Kubernetes Deplo...

Student Profile

Student Profile

Student Profile Rep...

Board

Microservices Sub...

Power

Board

Civil

No Records in Civil

CSE

James Wilson
STU-2026-003

Michael Chen
STU-2026-005

No Records in CSE

EEE

No Records in EEE

ECE

No Records in ECE

Container Assessment Form

Student Profile

Done

Basic Profile

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Time

Stop Down

Radio

Multi Select

Checkbox

Student Name

Register Number

Department

Submission Date

Assessment Id

26

Field Properties

Field name

Student Name

Field is a name

student_name

View Field References

Validation

Mandatory

No duplicate values

Initial value

Initial value

Character maximum

255

Data Privacy

Contains personal data (PII)

Data Security

Encrypted data

Contains health data (PHI)

User input for mobile

Full

QR code

Barcode